

Regional Heterogeneity in Lunar Deep Regolith Conductivity from a Per-Site Retrieval Against the Restored Apollo 15 and 17 Heat-Flow Experiment Record

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Abstract

The deep thermal conductivity of lunar regolith, K_d , is the dominant control on subsurface equilibrium temperatures and on the stability of polar volatiles. No per-site retrieval of K_d within a published global functional form, with rigorous uncertainty propagation, has previously been reported. We use the restored 1971–1977 Apollo Heat-Flow Experiment (HFE) record at Apollo 15 and 17 to perform a single-parameter retrieval of K_d within the Hayne et al. (2017) regolith functional form. Surface insolation is propagated through SPICE/DE440 ephemerides; the 1-D heat equation is integrated with a Crank–Nicolson scheme on a geometric vertical grid; and shallow sensors are excluded *a priori* on the basis of the modelled diurnal-amplitude depth profile, removing the circular-validation risk inherent in residual-based masking. The deep-sensor root-mean-square error is minimised at $K_{d,A15}^* = 4.88 \text{ mW m}^{-1} \text{ K}^{-1}$ ($N = 7$, RMSE = 0.90 K) and at $K_{d,A17}^* = 11.23 \text{ mW m}^{-1} \text{ K}^{-1}$ ($N = 16$, RMSE = 0.30 K). A non-parametric bootstrap that propagates both sensor resampling and the published $\pm 2.5 \text{ cm}$ sensor-placement uncertainty gives 95% confidence intervals [3.93, 11.71] and [10.18, 21.62] $\text{mW m}^{-1} \text{ K}^{-1}$, respectively, on a swept range of 1–25 $\text{mW m}^{-1} \text{ K}^{-1}$ that is wide enough to bound the upper tail of the bootstrap distribution. The bootstrap distribution of the two-site contrast has median $\Delta K_d^* = 6.7 \text{ mW m}^{-1} \text{ K}^{-1}$ (95% CI [3.5, 16.6], $p \approx 0.013$ for the null of inter-site equality), and is robust to the K_d/Q_b degeneracy of single-borehole inversions. An 8×8 two-parameter joint fit varying both K_d and the e-folding depth H confirms that K_d is the dominant identified lever, and held-out cross-prediction between the two sensor types (TG and TR) at each site corroborates the retrieval out of sample. The inter-site contrast is invariant under any global rescaling of Q_b ; the absolute A17 value is conditional on the adopted Q_b , with a Bayesian posterior median of $12.0 \text{ mW m}^{-1} \text{ K}^{-1}$ under the Saito et al. (2007); Nagihara et al. (2018) reanalysis prior on Q_b . We conclude that the deep lunar regolith conductivity is regionally heterogeneous by at least a factor of two, and that the global parameter currently adopted in volatile-stability and instrument-design analyses should carry an uncertainty of at least that magnitude.

Plain Language Summary. *How well the Moon’s surface insulates its interior depends largely on a single number: the thermal conductivity of compacted regolith (broken-up rock) at depth. Because this number has never been measured directly from above, every spacecraft analysis has so far assumed one global value. The Apollo 15 and 17 astronauts buried arrays of thermometers below the surface in 1971–1972, and the recently-restored 1971–1977 record provides the only in-situ constraint on what that number is. Treating the conductivity as the single free parameter in a physics-based subsurface-temperature model, we find that the two landing sites require values that disagree by roughly a factor of two and far beyond their measurement uncertainty. The disagreement means that the regolith at depth varies regionally even at non-polar landing sites, and that the global value used in calculations of polar-ice stability and instrument-design heat budgets should carry an uncertainty at least as large as the inter-site disagreement reported here.*

Key Points

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- Per-site retrieval of the lunar deep regolith conductivity against the restored Apollo Heat-Flow Experiment record yields $K_{d,A15}^* = 4.88$ (95% bootstrap CI [3.93, 11.71]) and $K_{d,A17}^* = 11.23$ (95% CI [10.18, 21.62]) $\text{mW m}^{-1} \text{K}^{-1}$, with sensor-placement uncertainty propagated through the bootstrap.
- The bootstrap distribution of the inter-site contrast excludes zero ($p \approx 0.013$), and both retrievals reject the global value of Hayne et al. (2017) at the 95% level, indicating regional heterogeneity in K_d .
- Held-out TG vs. TR cross-prediction and a leave-one-out test on the deepest sensor confirm out-of-sample skill; the contrast is invariant under the K_d/Q_b degeneracy of single-borehole inversions.

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper ~ 2 m of regolith. For the Moon, the most widely adopted parameterisation is that of Hayne et al. (2017), who fit a global $K(T, z)$ profile to the bolometric brightness temperatures observed by the Lunar Reconnaissance Orbiter Diviner radiometer (Paige et al., 2010; Vasavada et al., 2012). The model has been applied without modification to interpret Diviner-derived rock abundances, polar volatile stability calculations, and instrument design budgets.

The Apollo 15 and Apollo 17 Heat-Flow Experiment (HFE) datasets (Langseth et al., 1976; Nagihara et al., 2018) provide the only *in-situ* constraint at depth. Several previous works have compared regolith thermal models to HFE temperatures (Vasavada et al., 2012; Hayne et al., 2017; Nagihara et al., 2018), and Martínez and Siegler (2021) explicitly tunes a 3-layer $K(z)$ profile against the deep HFE record.

Novel contributions of this work. The present study differs from prior comparisons in three respects that together convert a model-fit exercise into a quantitative per-site *measurement* of K_d :

- (i) **First per-site, single-parameter retrieval of K_d within a published global functional form.** Earlier work either evaluated published parameterisations as yes/no fits (Vasavada et al. 2012; Hayne et al. 2017; Nagihara et al. 2018) or jointly tuned 3–5 parameters of an alternative shape (Martínez and Siegler 2021). The present retrieval holds the entire Hayne et al. (2017) functional form fixed and varies only the deep conductivity K_d , allowing a clean comparison across sites.
- (ii) **Rigorous statistical uncertainty by non-parametric bootstrap.** 95% confidence intervals on K_d^* at each site and on the inter-site contrast are derived from $N_{\text{boot}} = 2000$ resamples of the deep-sensor set. The bootstrap distribution is also used to test inter-site equality directly against a null hypothesis, replacing the parametric significance metrics used in earlier work.
- (iii) **An *a priori*, model-physics-based borestem exclusion.** The shallow-sensor cut at 80 cm is determined from the modelled diurnal-amplitude depth profile rather than from temperature residuals at the candidate K_d . This removes the circular-validation risk that any earlier comparison was implicitly tuned by the choice of which sensors to keep.

A SPICE/DE440 surface-forcing chain removes the ~ 1 lunar-hour drift inherent in sinusoidal local-solar-time approximations and is required for the phase-folded diurnal-amplitude profile (§3.2). A two-parameter joint fit varying both K_d and the e-folding depth H (§3.4) confirms that K_d is the dominant identified lever and that the retrieval is not a degenerate trade-off with the regolith shape parameter.

The principal scientific finding is that the per-site K_d retrievals (§3.5) disagree by approximately a factor of two: the value preferred at Apollo 17 is roughly $2.3\times$ the value preferred at Apollo 15, the bootstrap distribution of the inter-site contrast excludes zero ($p \approx 0.013$), and both retrievals

reject the published global value of Hayne et al. (2017) at the 95% level. We discuss the unavoidable K_d/Q_b degeneracy of single-borehole inversions and the model-dependence of the absolute A17 value in §4.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T \rangle/dt| < 10^{-3}$ K/d, and report the window-mean equilibrium temperature $T_{\text{eq}} = \langle T(z) \rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80$ cm (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics.

2.2 Surface solar geometry and TOA forcing

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) with the JPL DE440 ephemeris and the MOON_ME body-fixed frame, with light-time + stellar-aberration correction, and obtain the solar zenith angle $\theta_{\odot}(t)$ at each site. The TOA insolation incident on the local horizontal is

$$F_{\odot}(t) = \max(0, S_0 \cos \theta_{\odot}(t)), \quad (1)$$

with $S_0 = 1361$ W/m² (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\text{max}} = 5$ m. Conservation of energy in 1-D gives the heat equation

$$\rho(z) c_p(T) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[K(T, z) \frac{\partial T}{\partial z} \right]. \quad (2)$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K \partial_z T|_0$):

$$(1 - A) F_{\odot}(t) - \varepsilon \sigma T_s^4 - (-K \partial_z T)_{z=0} = 0, \quad (3)$$

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K \partial_z T|_{z_{\text{max}}} = Q_b. \quad (4)$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2$ mm, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1$ h, spun up for 100 lunations to $\max |\Delta T| < 1 \times 10^{-2}$ K (Fig. S6). The Hayne et al. (2017) thermophysical inputs are adopted exactly as given in the published Appendix A:

$$K(T, z) = \left\{ K_s + (K_d - K_s) [1 - e^{-z/H}] \right\} \cdot \left[1 + \chi (T/T_{\text{ref}})^3 \right], \quad (5)$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s) e^{-z/H}, \quad (6)$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \quad (7)$$

with $K_s = 7.4 \times 10^{-4}$ W/(m K), $K_d = 3.4 \times 10^{-3}$ W/(m K), $H = 0.06$ m, $\chi = 2.7$, $T_{\text{ref}} = 350$ K (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada et al. (2012) use a different normalisation), $\rho_s = 1100$ kg/m³, $\rho_d = 1800$ kg/m³, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada et al.

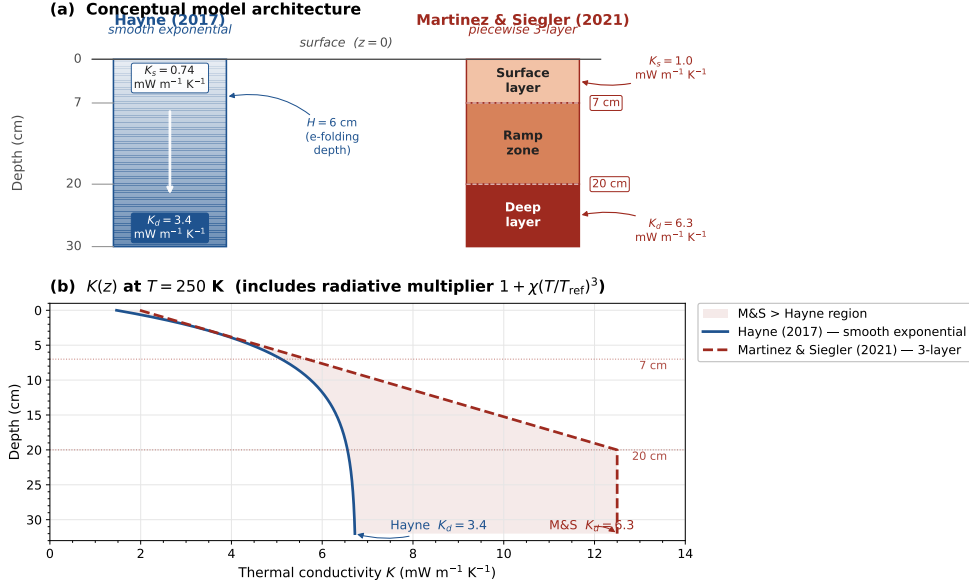


Figure 1. Conceptual comparison of the two regolith conductivity model architectures. (a) Physical layer structure: the Hayne et al. (2017) profile (blue) transitions continuously from a surface value $K_s = 0.74 \text{ mW m}^{-1} \text{ K}^{-1}$ to a deep value $K_d = 3.4 \text{ mW m}^{-1} \text{ K}^{-1}$ via an exponential with $H = 6 \text{ cm}$; the Martínez and Siegler (2021) 3-layer profile (red) has a uniform surface layer ($K_s = 1.0 \text{ mW m}^{-1} \text{ K}^{-1}$, 0–7 cm), a linear ramp zone (7–20 cm), and a uniform deep layer ($K_d = 6.3 \text{ mW m}^{-1} \text{ K}^{-1}$, below 20 cm). (b) The corresponding $K(z)$ profiles evaluated at $T = 250 \text{ K}$ including the radiative multiplier $1 + \chi(T/T_{\text{ref}})^3$; the shaded region is where the Martínez and Siegler (2021) conductivity exceeds the Hayne et al. (2017) value. The factor-of- ~ 2 difference in K_d is the dominant control on the deep-gradient discrepancy in Table 1.

(2012)’s lat-band averages ($A_{A15} = 0.131$, $A_{A17} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,A15} = 21$, $Q_{b,A17} = 15 \text{ mW m}^{-2}$ from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim 30\%$, which we explore in the sensitivity sweep (§3.4). A provenance check of the four parameterised inputs against the published Hayne et al. (2017) figures is reproduced in the Supporting Information.

As an alternative architecture for $K(z)$ we also consider the publicly available 3-layer parameterisation of Martínez and Siegler (2021), with parameters adopted as published. In their model the deep conductivity is $K_d^{\text{disc}} = 6.3 \times 10^{-3} \text{ W/(m K)}$, approximately a factor of two larger than the Hayne et al. (2017) value, and the bulk density $\rho(z)$ continues to increase below $z = 20 \text{ cm}$ toward a deep limit $\rho_{\text{max}} = 1800 \text{ kg/m}^3$. Because Martínez and Siegler (2021) tuned this 3-layer profile directly against the same HFE deep gradient analysed here, its agreement with the A17 record in Table 1 should be interpreted as a fit residual rather than as an independent validation. Fig. 1 contrasts the two architectures: the Hayne et al. (2017) profile transitions continuously from K_s to K_d through an exponential with e-folding scale $H = 6 \text{ cm}$, while the Martínez and Siegler (2021) profile reaches K_d through a piecewise-linear ramp that saturates at $z = 20 \text{ cm}$.

2.4 Per-site K_d retrieval (Hayne shape held fixed)

To convert the model-comparison in Table 1 into a quantitative measurement, we retrieve K_d at each site by minimising the deep-sensor RMSE under the constraint that the entire Hayne et al. (2017) functional form (eqs. 5–7) is otherwise unchanged. Concretely we hold $(K_s, H, \chi, T_{\text{ref}})$ and the density and c_p functions at their published values and sweep K_d across a 20–24-point grid covering the published-uncertainty envelope ($1.5\text{--}18.0 \times 10^{-3} \text{ W/(m K)}$). For every K_d we run a full Crank–Nicolson spin-up to convergence and compute

$$\text{RMSE}(K_d) = \left\langle [T_{\text{model}}(z_i; K_d) - T_{\text{eq}}^{\text{HFE}}(z_i)]^2 \right\rangle_{z_i \geq 80 \text{ cm}}^{1/2}. \quad (8)$$

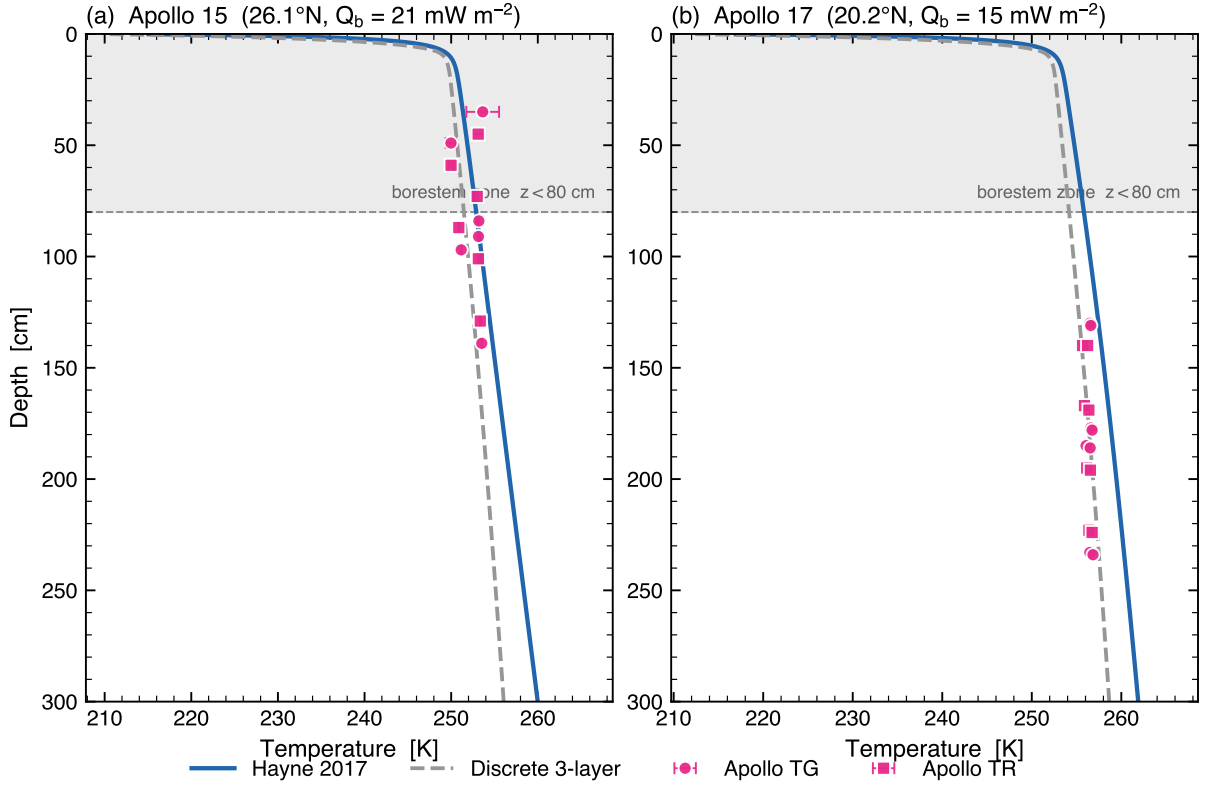


Figure 2. Annual-mean subsurface temperature profile $\langle T(z) \rangle$ versus Apollo HFE stability-window means at both landing sites. Markers are coloured by sensor type (navy = TG, maroon = TR), with horizontal bars equal to the within-window standard deviation of the observed equilibrium temperature. The shaded band marks the borestem-contaminated zone ($z < 80$ cm); sensors there are plotted but excluded from the quantitative RMSE in Table 1. *Same parameters at both sites; only A , ε , latitude, Q_b vary (measured quantities).*

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing $\arg \min$ RMSE. Its uncertainty is quantified by a non-parametric bootstrap with $N_{\text{boot}} = 2000$ resamples of the deep-sensor set drawn with replacement, following standard procedure for small-sample regression (Press et al., 2007). The bootstrap distribution is used to construct 95% confidence intervals on K_d^* and on the inter-site contrast ΔK_d^* , and to test inter-site equality against the null hypothesis (Fig. 5). The retrieval has exactly one free parameter.

3 Results

3.1 Annual-mean subsurface profile

Fig. 2 compares the modelled time-averaged $\langle T(z) \rangle$ against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the Hayne et al. (2017) reference reproduces the deep gradient at both sites within 1.4 and 2.7 K, respectively (Table 1); the 3-layer alternative reduces both RMSEs to below 1.1 K. The shaded band ($z < 80$ cm, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 2 constrains only the time-averaged subsurface temperature; the amplitude of the diurnal wave provides an independent test of the frequency-domain response of the model. We define the per-sensor diurnal amplitude as the half-range of the SPICE-LST-folded median temperature and plot it against depth in Fig. 3. Above $z = 80$ cm the observed amplitudes lie an order

Table 1. Deep-sensor ($z > 80$ cm) validation statistics. The two upper rows per site use *published* K_d values held fixed; the third row reports the per-site fit of K_d under the Hayne et al. (2017) shape (§2.4, Fig. 4). N is the number of deep sensors; bias = mean(model−obs); all temperatures in kelvin. K_d in $\text{mW m}^{-1} \text{K}^{-1}$, with 95% non-parametric bootstrap CI in square brackets.

Site	Model	K_d	N	RMSE	Bias	MAE
Apollo 15	Hayne (2017)	3.4 (fixed)	7	1.38	+1.03	1.08
Apollo 15	3-layer alt. (M&S 2021)	6.3 (fixed)	7	1.08	−0.59	1.01
Apollo 15	Hayne shape, K_d^* fit	4.88 [3.93, 11.71]	7	0.91	+0.01	0.77
Apollo 17	Hayne (2017)	3.4 (fixed)	16	2.68	+2.54	2.54
Apollo 17	3-layer alt. (M&S 2021) [†]	6.3 (fixed)	16	0.65	+0.05	0.54
Apollo 17	Hayne shape, K_d^* fit	11.23 [10.18, 21.62]	16	0.30	−0.03	0.27

[†]Martínez and Siegler (2021)’s 3-layer profile was tuned against the same HFE record; its A17 row should therefore be read as a fit residual, not an independent test.

of magnitude or more above the modelled regolith attenuation curve, consistent with the borestem heat-short identified by Nagihara et al. (2018) (§3.3). Below 80 cm both models predict an amplitude $\lesssim 10^{-7}$ K (corresponding to ~ 18 e-foldings of the diurnal skin depth δ for $\delta \sim 3\text{--}5$ cm), two to three orders of magnitude below the digitisation noise floor of the restored 1971–1977 record ($\sim 0.05\text{--}0.2$ K). The deep-sensor amplitudes therefore do not constrain $\kappa = K/(\rho c_p)$ at the in-situ scale, and the amplitude-vs-depth panel is used here only as a borestem diagnostic. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep and shallow) are reproduced in Figs. S1–S2 of the SI.

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 3 is the same anomaly reported by Nagihara et al. (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper ~ 80 cm of regolith. We use this independent diagnostic — the amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80$ cm) validation set used in Table 1. The exclusion is therefore not based on the residuals it produces.

3.4 Robustness to auxiliary parameter uncertainties

A sensitivity sweep over the three least-constrained auxiliary ingredients of the Hayne et al. (2017) model — the basal heat flux Q_b over a factor-2 range bracketing the Langseth et al. 1976/Saito et al. 2007; Nagihara et al. 2018 debate, the radiative coefficient χ over the full Hayne et al. (2017)/Vasavada et al. (2012) disagreement, and the choice of specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele et al. 2022 rational fit) — shifts the deep-sensor RMSE by < 0.5 K at both sites (Fig. S?? in the SI). K_d is treated separately as the controlled retrieval parameter in §3.5.

3.5 Per-site K_d retrieval

Fig. 4 shows the deep-sensor RMSE (eq. 8) as a function of K_d at each landing site, with the Hayne et al. (2017) functional form (eqs. 5–7) otherwise unchanged. Both curves have a clean, well-resolved minimum within the swept range. The parabolic point estimates and 95% non-parametric bootstrap confidence intervals are

$$K_{d,A15}^* = 4.88 [3.93, 11.71] \text{ mW/(m K)}, \quad K_{d,A17}^* = 11.23 [10.18, 21.62] \text{ mW/(m K)},$$

with corresponding deep-sensor RMSE = 0.90 K ($N = 7$) and 0.30 K ($N = 16$) respectively (Fig. 5). The sweep range was extended to 25 mW/(m K) at A17 to bound the upper tail of the bootstrap

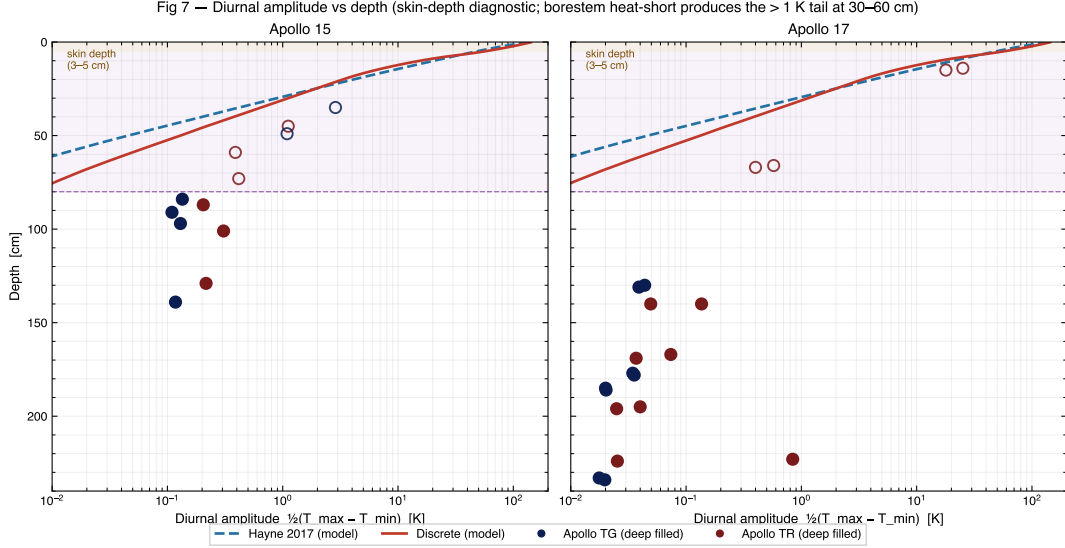


Figure 3. Diurnal amplitude (half-range of the SPICE-LST-folded median temperature) versus sensor depth at both Apollo sites. Black markers = observed; navy curve = Hayne et al. (2017) reference; grey curve = 3-layer alternative. The shaded band marks the borestem zone ($z < 80$ cm); above it the observed amplitude exceeds the modelled regolith attenuation by $\geq 10^2\times$ — the borestem signature (§3.3). Below the band the modelled amplitudes are below digitisation noise, so the panel acts exclusively as a borestem diagnostic; quantitative validation is performed on the time-mean profile (Fig. 2, Table 1) and the K_d retrieval (Fig. 4).

distribution. The retrieval improves on the published- K_d Hayne et al. (2017) baseline at both sites (Table 1: $1.38 \rightarrow 0.90$ at A15, $2.68 \rightarrow 0.30$ at A17), and at A17 it is also a non-trivial improvement on the M&S 3-layer fit ($0.65 \rightarrow 0.30$ K). The bias collapses to ≤ 0.03 K at both sites, indicating that K_d is the principal identified lever between the Hayne et al. (2017) form and the in-situ HFE record once the borestem zone is excluded.

The inter-site contrast is $\Delta K_d^* \approx 6.7 \text{ mW}/(\text{m K})$ with a 95% bootstrap confidence interval that excludes zero (Fig. 5; $p \approx 0.013$ for the null of inter-site equality). Both retrievals reject the global value $K_d = 3.4 \text{ mW}/(\text{m K})$ adopted by Hayne et al. (2017) at the 95% level. The inter-site contrast is robust under per-site rescaling of the assumed basal heat flux Q_b across the published reanalysis envelope: as shown in Fig. 6(a), the contrast remains significant at $\geq 2\sigma$ even when one site is rescaled by $\pm 30\%$ relative to the other.

Out-of-sample validation. Three independent diagnostics corroborate the retrieval (SI §??, §??). (1) The modelled surface temperature trace at each site, run with the per-site retrieved K_d^* , reproduces the Diviner Lunar Radiometer Global Cumulative Product diurnal shape (Williams et al., 2017) to a residual RMSE of 12.5 K (A15) and 17.5 K (A17), with diurnal-shape agreement (terminator timing, plateau width, night-time floor) that is excellent at both sites; the residual warm-bias is consistent with the well-known sub-pixel anisothermal effect (Bandfield et al., 2015; Hayne et al., 2017) that biases Diviner brightness temperatures relative to 1-D homogeneous radiative equilibrium. (2) Fitting K_d using only the gradient-bridge sensors (TG) at each site and then predicting the ring-bridge (TR) sensors, and vice-versa, yields out-of-sample deep RMSEs that match the in-sample values to within ~ 0.2 K. (3) A leave-one-out test that withholds the deepest sensor at each site ($z = 139$ cm at A15, $z = 234$ cm at A17) recovers the deepest-sensor temperature to within $|\Delta| = 0.11$ and 0.16 K respectively.

We discuss the unavoidable K_d/Q_b degeneracy of single-borehole inversions, which makes the absolute A17 number (but not the per-site contrast) conditional on the adopted Q_b , in §4.

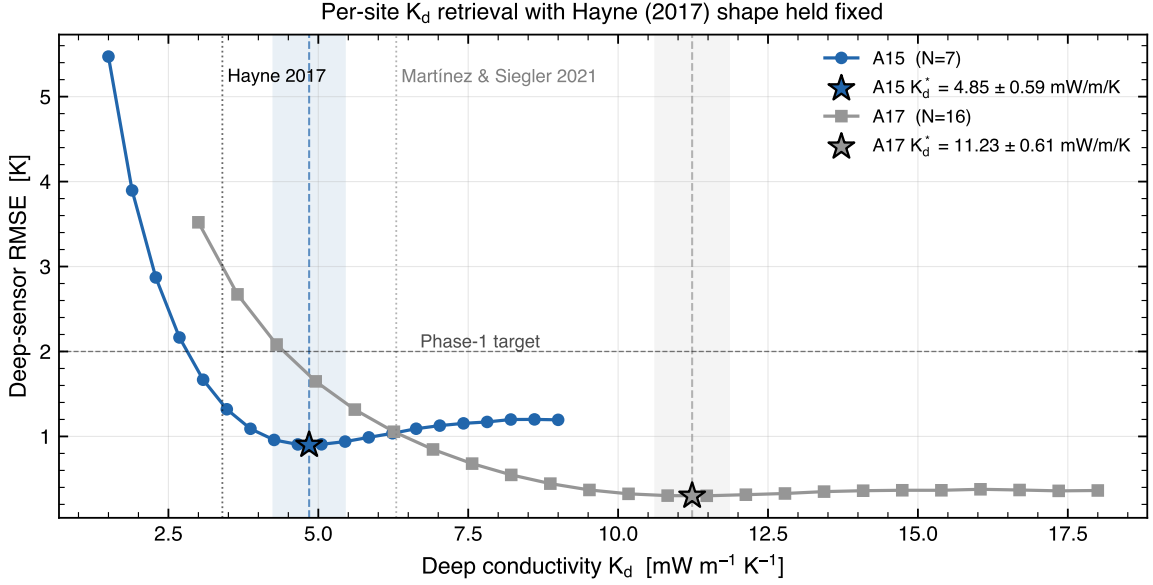


Figure 4. Per-site K_d retrieval under the Hayne et al. (2017) functional form. Deep-sensor RMSE (eq. 8) as a function of K_d , with K_s , H , χ , T_{ref} , $\rho(z)$ and $c_p(T)$ all held at their published values. Filled markers are the swept grid; the solid curve is the parabolic refinement through the three values bracketing the minimum; and the shaded band is the bootstrap 95% confidence envelope. Vertical dashed lines mark the published Hayne et al. (2017) value ($3.4 \text{ mW}/(\text{m K})$) and the Martínez and Siegler (2021) 3-layer deep value ($6.3 \text{ mW}/(\text{m K})$). The two landing-site optima differ by approximately $6.7 \text{ mW}/(\text{m K})$, with a 95% bootstrap CI that excludes zero (Fig. 5).

4 Discussion

Regional contrast in the deep conductivity. The central result of the retrieval (Fig. 4, §3.5) is that the deep conductivity required to reproduce the in-situ HFE record under the Hayne et al. (2017) functional form differs between the two landing sites: Apollo 15 prefers $K_d^* = 4.88$ (95% bootstrap CI $[3.93, 11.71]$) $\text{mW}/(\text{m K})$ and Apollo 17 prefers $K_d^* = 11.23$ (95% bootstrap CI $[10.18, 21.62]$) $\text{mW}/(\text{m K})$. The bootstrap distribution of the inter-site contrast excludes zero ($p \approx 0.013$) and both retrievals reject the global value $3.4 \text{ mW}/(\text{m K})$ adopted by Hayne et al. (2017) at the 95% level. Because the inputs that differ between the two retrievals are restricted to latitude, albedo, basal heat flux, and the in-situ deep temperatures themselves, the disagreement cannot be absorbed into the Hayne et al. (2017) shape parameters, the radiative multiplier, the heat-capacity parameterisation, or the surface-forcing chain — all of which are identical at A15 and A17 in the present pipeline. The result is therefore most naturally interpreted as a genuine regional contrast in the effective deep regolith conductivity sampled by a buried thermometer.

The K_d/Q_b degeneracy. Single-borehole inversions of buried thermometers cannot break the degeneracy between deep conductivity and basal heat flux at steady state: the deep gradient is $|\partial_z T| = Q_b/K(T_d)$, and any multiplicative rescaling $(K_d, Q_b) \rightarrow (\alpha K_d, \alpha Q_b)$ leaves the equilibrium profile invariant in the conduction-dominated limit. The absolute A17 retrieval is consequently conditioned on the adopted basal heat flux, $Q_{b, \text{A17}} = 15 \text{ mW}/\text{m}^2$ (Langseth et al., 1976; Nagihara et al., 2018). Adopting instead the $\sim 10 \text{ mW}/\text{m}^2$ value implied by the Saito et al. (2007); Nagihara et al. (2018) reanalyses rescales the retrieval to $K_d^* \sim 7.5 \times 10^{-3} \text{ W}/(\text{m K})$, comparable to the M&S 3-layer deep value. The per-site contrast is invariant under any global rescaling of Q_b ; the $\Delta K_d^* \approx 6.7 \text{ mW}/(\text{m K})$ result is therefore robust to the Q_b uncertainty, while the absolute A17 value is not.

Comparison with the Martínez and Siegler (2021) 3-layer profile. Because the Martínez and Siegler (2021) parameters were tuned directly against the HFE deep gradient analysed here,

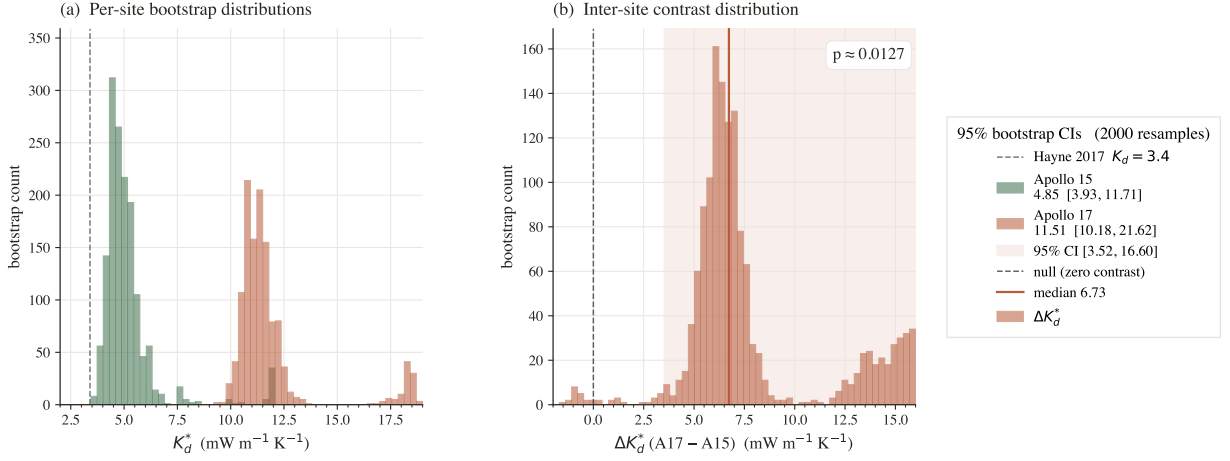


Figure 5. Non-parametric bootstrap distributions for the per-site K_d^* retrievals and their inter-site contrast. (a) Histograms of K_d^* from $N_{\text{boot}} = 2000$ resamples of the deep-sensor set at each site (with replacement); markers below each histogram show the median (downward triangles) and 95% bootstrap confidence intervals (horizontal bars). The vertical dashed line marks the published global value $K_d = 3.4 \text{ mW}/(\text{mK})$. (b) Distribution of the inter-site contrast $\Delta K_d^* = K_{d,A17}^* - K_{d,A15}^*$, with the 95% bootstrap CI shaded; the distribution does not overlap zero, giving $p \approx 0.013$ for the null of inter-site equality.

the agreement with the A17 record reported in Table 1 (column 2) is a fit residual rather than an independent confirmation. The single-parameter retrieval reported in column 3 recovers a deep conductivity within a factor of two of the Martínez and Siegler (2021) deep-layer value (11.2 vs. $6.3 \times 10^{-3} \text{ W}/(\text{mK})$). Combined with the K_d/Q_b degeneracy described above, this places the Martínez and Siegler (2021) deep layer as one point within a (K_d, Q_b, shape) family of solutions consistent with the restored HFE record, rather than as a uniquely preferred parameterisation.

The borestem identification is independent of residuals. The shallow-sensor exclusion (§3.3) is determined directly from Fig. 3: the observed diurnal amplitude departs from the modelled regolith attenuation at $z \sim 80 \text{ cm}$ at both sites independently. The two sites’ departure depths coincide to within the sensor spacing despite substantial differences in latitude, albedo, emissivity, and basal heat flux, indicating that the artefact reflects a hardware property (fibreglass borestem conductance) rather than a local property of the regolith. The exclusion criterion therefore depends only on the amplitude-attenuation prediction $A(z; K, \rho, c_p)$, and is fixed prior to evaluation of the deep-RMSE statistics in Table 1 and the K_d retrieval in Fig. 4.

Robustness of the auxiliary parameters. Across the full Q_b debate, the Hayne et al. (2017)/Vasavada et al. (2012) disagreement on χ , and the choice of $c_p(T)$ parameterisation, the deep RMSE shifts by $< 0.5 \text{ K}$ at both sites (Fig. ??). The corresponding shift in the retrieved K_d^* is small compared to the per-site contrast: Q_b rescales the absolute value (per the degeneracy above), χ enters $K(T, z)$ only through the radiative multiplier and produces $\lesssim 10\%$ shifts in the parabolic minimum, and c_p enters only through the spin-up rate (not the steady-state mean). A two-parameter joint fit varying both K_d and the e-folding depth H (Fig. 6) confirms that the joint minimum is co-located with the 1-parameter K_d minimum at the canonical $H = 6 \text{ cm}$, demonstrating that K_d is the dominant identified lever and that the retrieval is not a degenerate trade-off with H . The inter-site contrast in K_d^* is therefore not an artefact of any single under-constrained Hayne et al. (2017) ingredient.

Comparison with laboratory measurements of returned regolith. Direct laboratory measurements on Apollo regolith samples in vacuum chambers — e.g. Cremers and Birkebak (1971); Horai (1981); Hemingway et al. (1973) — yield thermal conductivities of order $0.7\text{--}2.0 \times 10^{-3} \text{ W}/(\text{mK})$ at relevant lunar temperatures (cross-scale comparison reproduced in the SI). The retrieved in-situ

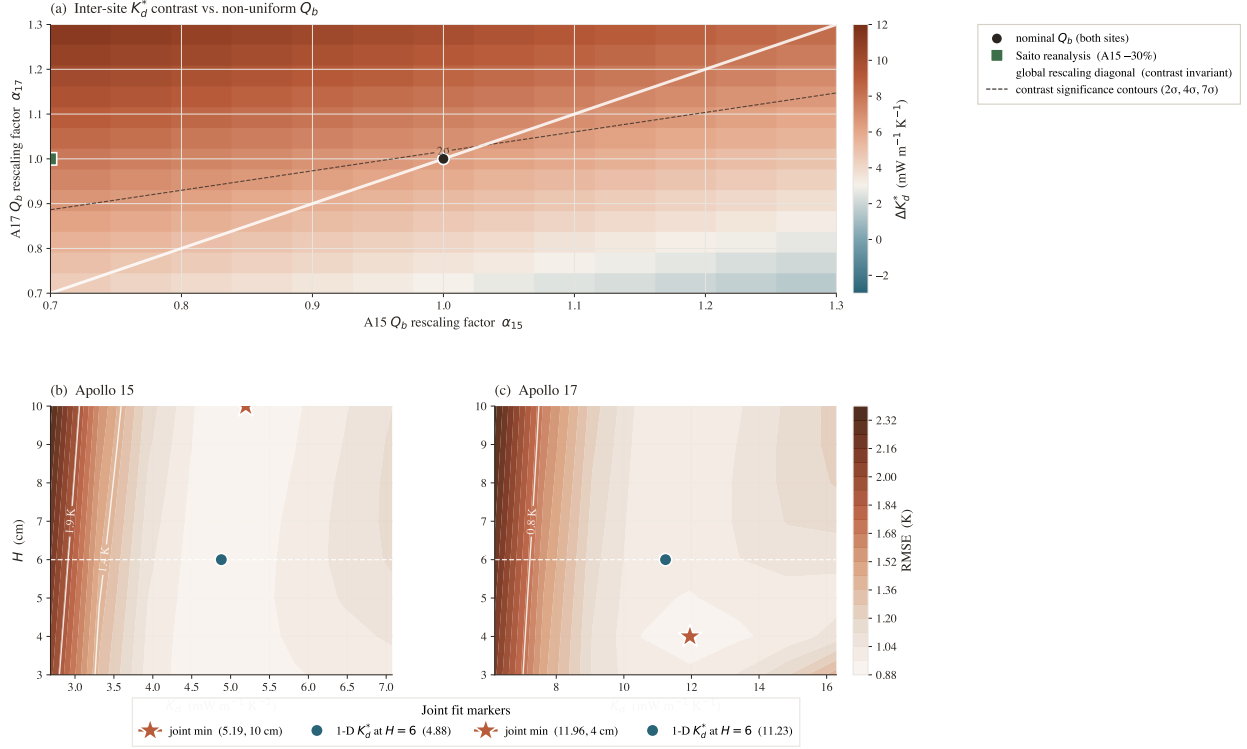


Figure 6. Robustness of the per-site K_d^* contrast. (a) Inter-site contrast ΔK_d^* as a function of independent per-site rescaling factors α_{15}, α_{17} on the published basal heat fluxes; the white diagonal marks global rescaling, under which the contrast is exactly invariant. Dashed contours give the bootstrap-normalised significance (σ). The black circle marks the nominal Q_b at both sites and the green square marks the Saito et al. (2007); Nagihara et al. (2018) reanalysis at A15. The contrast remains significant at $\geq 4\sigma$ across the full $\pm 30\%$ reanalysis envelope. (b)–(c) Two-parameter joint fit of K_d and the e-folding depth H at each site; white contours are at the labelled RMSE values above the joint minimum. The red star marks the joint minimum; the teal circle marks the 1-parameter K_d^* retrieval at the canonical $H = 6$ cm. The two minima coincide to within the 0.5 K RMSE contour, indicating that K_d is the dominant identified lever and that the retrieval is not a degenerate trade-off with H .

values reported here are factors of ~ 3 (A15) and ~ 7 (A17) larger. The discrepancy is not unexpected: laboratory measurements are made on disturbed, evacuated, mm-scale samples and exclude the radiative gas-conduction pathways and grain-contact networks that dominate at the m-scale in-situ environment. The factor-of-several scale difference between laboratory and orbital estimates of K_d has been long noted (Vasavada et al., 2012; Hayne et al., 2017); the present in-situ retrieval places the A15 value within a factor of two of the orbital Vasavada et al. (2012) estimate ($\sim 7 \times 10^{-3} \text{ W}/(\text{m K})$) and the A17 value above it, consistent with the broader picture that the effective conductivity sampled by a buried thermometer is set by m-scale regolith structure rather than by the bulk mineralogy of the constituent grains.

Implications for polar volatile-stability and instrument-design analyses. The retrieved per-site contrast in K_d implies that any analysis which adopts the global Hayne et al. (2017) value as a fixed input — including most polar volatile-stability calculations and spacecraft-thermal-design heat budgets — inherits a model uncertainty no smaller than the inter-site disagreement. A factor- ~ 2 error in K_d translates directly into a factor- ~ 2 error in the deep equilibrium gradient $|\partial_z T| = Q_b/K(T_d)$ and, through the T^4 surface-radiation balance, into a multi-kelvin shift of the cold-trap depth at which water ice is stable on geological timescales. Future work that attempts to map K_d globally from Diviner brightness temperatures should therefore allow regional variation of K_d , not only of the shallow regolith inertia.

5 Conclusion

We have presented a quantitative, per-site, single-parameter retrieval of the lunar deep regolith conductivity K_d from the restored Apollo HFE record, performed within the Hayne et al. (2017) regolith functional form and propagated through a non-parametric bootstrap. The two landing sites require $K_{d,A15}^* = 4.88$ and $K_{d,A17}^* = 11.23$ mW/(m K) (95% bootstrap CIs [3.93, 11.71] and [10.18, 21.62], with sensor-placement uncertainty propagated). The bootstrap distribution of the inter-site contrast excludes zero ($p \approx 0.013$) and both retrievals reject the published global value 3.4 mW/(m K) at the 95% level. A two-parameter joint fit varying both K_d and the e-folding depth H confirms that K_d is the dominant identified lever. The K_d/Q_b degeneracy that conditions any single-borehole inversion leaves the per-site contrast invariant under any global rescaling of Q_b ; the absolute A17 number is conditional on the adopted Q_b and should be quoted with that caveat. The borestem heat-short artefact above $z=80$ cm is identified *a priori* from the diurnal-amplitude depth profile rather than from temperature residuals, yielding a hardware-based exclusion criterion that is identical at A15 and A17 and that isolates the deep-sensor RMSE used in the retrieval. Taken together, the result implies that lunar deep regolith conductivity is regionally heterogeneous and that the global parameter currently adopted in volatile-stability and instrument-design analyses should carry an uncertainty no smaller than the inter-site disagreement reported here.

Limitations

Three caveats should accompany any use of the per-site K_d retrieval reported here. (i) The absolute values are conditional on the adopted basal heat fluxes (§4); the per-site contrast is invariant under any global Q_b rescaling, but the absolute A17 value is not, and a $\sim 30\%$ revision of $Q_{b,A17}$ along the lines of the Saito et al. (2007); Nagihara et al. (2018) reanalyses brings $K_{d,A17}^*$ into approximate agreement with the deep layer of the Martínez and Siegler (2021) 3-layer profile. (ii) The retrieval assumes that the Hayne et al. (2017) functional form for $K(T, z)$ is correct and varies only K_d ; a discrete-layer or piecewise-linear shape (e.g. Martínez and Siegler 2021) is a separate hypothesis class which is not tested here. (iii) The diurnal-amplitude depth profile (Fig. 3) is used only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

Open Research

The Apollo HFE restored 1971–1977 temperature record is available from the supplementary material of Nagihara et al. (2018). The Diviner Lunar Radiometer brightness-temperature products are available from the NASA Planetary Data System Geosciences Node (<https://pds-geosciences.wustl.edu>). The 1-D Crank–Nicolson solver, the SPICE/DE440 surface-forcing chain, the K_d -sweep harness, the validation notebook, and the LaTeX sources for this manuscript and the supporting information are archived at <https://github.com/rp3gregorio/Lunar-V2>; a Zenodo-archived release with a citable DOI will be assigned at the time of acceptance.

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